

CardioAI - Application Architecture & Tech Stack

1. Executive Summary & Application Overview

CardioAI (CardioProgress) is a dual-platform medical AI system designed for cardiac fibrosis risk assessment, biomarker tracking, and report diagnosis. It operates seamlessly across both Android Mobile Devices and Web Browsers connected to a shared cloud database.

2. Frontend Technologies & Languages

A. Android Mobile App (kotlinapp):

- ? Programming Language: Kotlin (1.9+)
- ? UI Framework: Jetpack Compose (Declarative Modern UI Framework)
- ? Design Tokens & Components: Material 3, Custom Card Gradients, Canvas Gauges
- ? Concurrency: Kotlin Coroutines & StateFlow (Dispatchers.IO, ViewModel lifecycle)
- ? Local Persistence: SharedPreferences (AppSettings persistent session manager)

B. Web Application (webapp):

- ? Markup Language: HTML5 (Semantic Structure, Accessible Form Elements)
- ? Styling & Aesthetics: Vanilla CSS3 (Glassmorphism, Dark/Light Themes, Flexbox/Grid)
- ? Client Logic: JavaScript (ES6+ Modern Syntax, Asynchronous Promises, Event Engine)
- ? Icons & Typography: FontAwesome 6 Pro, Google Inter/Outfit Fonts
- ? Local Cache: HTML5 LocalStorage (Instant zero-latency UI rendering)

3. Backend Technologies, Cloud DB & AI Engine

A. Database & Cloud Architecture:

- ? Database Engine: Firebase Firestore (NoSQL Document-based Cloud Database)
- ? Collections Schema: 'users' (Profiles), 'patient_details' (Vitals/BMI), 'reports' (Scans)
- ? User Authentication: Firebase Auth (Email/Password Auth, UID-based Security)

B. Artificial Intelligence & Optical OCR Engine:

- ? Document Reader: Google ML Kit Vision OCR (Text Recognition v2)
- ? Generative AI: Google Gemini AI API (LLM Biomarker Summarization & Clinical Insights)
- ? Heuristic Risk Classifier: Evaluates Troponin I, BNP, NT-proBNP & LVEF metrics

4. Real-Time Data Synchronization Architecture

- ? Android Listener: DisposableEffect binding addSnapshotListener on Firestore documents.
- ? Web Listener: Firebase Web SDK onSnapshot observers attached to user & patient collections.
- ? Multi-Device Sync: Any change made on Web updates Android live within 1 second and vice versa.
- ? Data Normalization: Biomarker height_cm & weight_kg are stored consistently as Numbers.